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Tutorial: Large Language-Vision Model in Society

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Abstract

The tutorial "Large Vision-Language Model in the Society" aims to provide a comprehensive overview of state-of-the-art techniques and applications of large vision-language models (LVLMs), which integrate visual and textual data to transform multimedia research and applications. LVLMs are poised to revolutionize domains such as content creation, social media analysis, education, healthcare, and entertainment by enabling sophisticated content analysis, retrieval, and generation. This tutorial will cover the fundamentals of vision-language integration, state-of-the-art models, training techniques, applications, ethical considerations, and future directions. It is designed to be educational and instructive, providing an in-depth introduction rather than a cursory survey. Attendees will gain practical skills, and insights into the latest research, and engage in interactive sessions to reinforce learning. By addressing both technical and societal aspects, the tutorial will significantly benefit the multimedia community, driving innovation and progress in the field.

CCS Concepts

• **Computing methodologies** → **Computer vision**; **Natural language processing**.

Keywords

Large Vision-Language Model, visual and textual data, transform multimedia

ACM Reference Format:
Kaicheng Yu, Zhuang Shao, Siyuan Qi, and Dongfang Liu. 2024. Tutorial: Large Language-Vision Model in Society. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24), October 28-November 1, 2024, Melbourne, VIC, Australia.* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/3664647.3689175

1 Proposers

We list our team here:

- Kaicheng Yu, Assistant Professor, Westlake University.
- Zhuang Shao, Lecturer (Assistant Professor), Newcastle University.
- Siyuan Qi, Research Scientist, Beijing Institute for General Artificial Intelligence.

MM '24, October 28-November 1, 2024, Melbourne, VIC, Australia

ACM ISBN 979-8-4007-0686-8/24/10
https://doi.org/10.1145/3664647.3689175

2 Bio

Dr. Kaicheng Yu is an Assistant Professor at Westlake University, and is the principal investigator of the Autonomous Intelligence Laboratory (AutoLab). He obtained the Bachelor degree from the University of Hong Kong with first class honor and proceeded directly to a Ph.D. program at École Polytechnique Fédérale de Lausanne (EPFL) in Switzerland. His research interests encompass automated machine learning, computer vision, 3D perception, autonomous driving, with over 20 publications in prestigious international journals and conferences. In 2019, he was awarded the Qualcomm Innovation Fellowship (one of only four recipients in Europe that year), and in 2021, he joined DAMO Academy's Autonomous Driving Lab through the "Alibaba Star" program. In 2023 and 2024. Prior to returning to China, he is a member of Intel Intelligent Systems Lab and Abacus.AI. Additionally, he established KMind as a co-founder and Chief Scientist, a company that secured tens of millions in investment and was awarded in 2023 as "Key Recommendation for Leading Project in Yuhang Hangzhou."

Dr. Zhuang Shao is a Lecturer (Assistant Professor) in Data Engineering and AI with the School of Engineering, Newcastle University. He obtained his PhD at Warwick Manufacturing Group (WMG), University of Warwick. His research interests are computer vision and machine learning, specifically vision-language learning, multimedia data captioning, and pedestrian detection. He has published more than 10+ high-quality papers, including ACM Multimedia, T-NNLS, T-MM, T-CSVT and T-ITS. Due to his excellent research contributions, he was awarded Star Awards in 2022 and 2023 by WMG, as well as the Early Career Researcher Award by the Association of British Chinese Professors in 2022. Google Scholar.

Dr. Siyuan Qi is a research scientist and team lead at Beijing Institute for General Artificial Intelligence (BIGAI). He obtained his Ph.D. degree in Computer Science from UCLA in 2019 and worked at Google from 2019-2022. His research interest mainly focuses on building AI agents to solve complex tasks in open and multi-agent environments. Dr. Qi has published over 30 research papers in top conferences and journals, including ICLR, ICML, and Science Robotics, and **he has organized workshops at conferences such as CVPR**.[1 2]

3 Schedule

We aim to conduct a comprehensive half-day tutorial that will consist of three in-depth courses, each designed to cover critical aspects of large vision-language models and their applications. These courses will provide participants with a thorough understanding of the state-of-the-art techniques, practical skills, and latest research developments in this rapidly evolving field. In addition to the structured courses, we will include an open discussion session to provide an opportunity for attendees to exchange ideas, ask questions, and engage in meaningful dialogue with the speakers and fellow participants. This interactive component is intended to foster collaboration, spark new insights, and address specific interests and challenges faced by the multimedia community.

8:40 am - 9:20 am, Course Talk 1
9:20 am - 9:30 am, Coffee break
9:30 am - 10:10 am, Course Talk 2
10:10 am - 10:20 am, Coffee break
10:30 am - 11:10 am, Course Talk 3
11:10 am - 12:00 am Open discussion

4 Courses

- **Course Talk 1**: *Can MLLMs Mirror Human Intelligence? An Evaluation Through the Lens of Human Cognitive Science*
  As recent multimodal large language models (MLLMs) have demonstrated impressive proficiency across a range of complex tasks, there has been growing debate about whether these models could eventually mirror human intelligence. However, existing benchmarks primarily focus on evaluating task performance, such as the accuracy of identifying an object's attributes. In this tutorial, we will begin by analyzing the limitations of traditional benchmarks. We will then provide an overview of efforts to incorporate psychological research into the evaluation of LLMs and trace the development history of human intelligence models. Following this, we will introduce our latest cognition-inspired multimodal benchmark, M3GIA, which leverages the well-established Cattell-Horn-Carroll (CHC) theory to understand the intelligence of MLLMs beyond superficial achievements. Finally, we will conclude with a Q&A session with the audience.
- **Course Talk 2**: *Dense Image Captioning for Comprehensive Scene*
  Interpretation: Deep Learning Approaches and Insights with Large Vision-Language Models. Brief Outline: Over the last decade, the exponential growth of image data has posed a significant challenge for human annotation. To address this, generic image captioning is always insufficient in capturing more comprehensive and intricate semantic content. In this tutorial, we will first introduce the motivations and basic ideas behind the dense image captioning task, explaining its necessity compared to generic captioning. Following this, we will delve into our research methodologies on dense image captioning, providing a detailed overview of these approaches and their effectiveness. After these, we are going to elaborately present its interactions with large vision language models and the potential challenges for these interactions with some interesting future works and applications followed by the QA session with the audience.
- **Course Talk 3**: *Building Agents for Open-Ended Environments*
  How to build agents that can solve open-ended complex tasks like humans? This talk explores the challenges and approaches in creating decision-making agents that can navigate through the intricacies of human society. We identify three critical aspects that need improvement for machine decision-making: efficient learning, effective reasoning, and ethical alignment. To address these challenges, we have developed an interactive environment called CivRealm. CivRealm is inspired by the Civilization game, where agents act as leaders guiding a civilization through eras and technological advancements. It poses new challenges in both learning and reasoning, encompassing mini-games and support for different agent types like reinforcement learning (RL) and large language models (LLM). To facilitate efficient learning, we introduce transformer-based models. These models can process complex inputs and have shown promising results in learning tasks. We also explore effective reasoning methods for LLM-based agents, enabling them to navigate complex scenarios in CivRealm. Lastly, we discuss multi-objective Pareto alignment for LLMs, which aims to align the model's objectives with ethical considerations.

5 Anticipated Audience

We aim to invite a diverse group of scholars from both academia and industry, along with their students and interns, to participate in this tutorial. This initiative seeks to foster collaboration and knowledge exchange between experienced professionals and emerging talents in the field of multimedia research. The anticipated participants include renowned experts, researchers, and practitioners who have made significant contributions to the development and application of large vision-language models. Additionally, we expect a strong attendance from graduate students and interns who are eager to learn and engage with cutting-edge advancements. By bringing together this mix of participants, we foresee creating a dynamic and enriching environment for all attendees. We anticipate having between 20-30 participants, ensuring a focused and interactive experience for everyone involved.

Dr. Steven C. H. Hoi, Managing Director of Salesforce Research Asia
Dr. David J. Crandall, Indiana University
Dr. Luc Van Gool, ETH Zurich
Dr. Thomas Tang, Nvidia
Dr. Yiming Cui, ByteDance
Dr. Zhiwen Cao, Adobe Research
Dr. Junhan Zhao, Harvard University

[1]https://scene-understanding.com/2019/index.html
[2]https://scene-understanding.com/2020/index.html